

Thinner Adamo

Dell hints at an even thinner version of its Adamo notebook at less than 10 mm

Paid movies on YouTube

YouTube talks to studios for streaming paid movies over the internet

Motherboard test



Jetway HA07 Ultra

the processor and the graphics card but the motherboard clearly wasn't a bottleneck anywhere. The scores seem a tiny more consistent than the motherboards in the Intel LGA775. Once again, there's little to comment on the performance differences.

AMD Socket AM2+/AM3 (Integrated graphics)

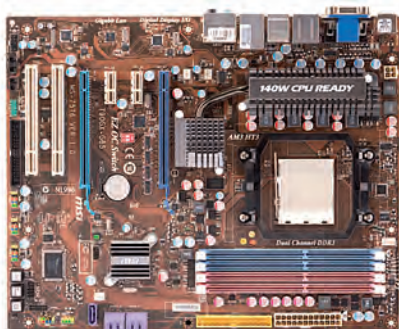
This category of the motherboards is dominated mainly by the slightly older but very popular AMD 780G chipsets and now the AMD 785G chipset. The NVIDIA 8xxx series of chipsets has been around but we haven't seen a lot of users adopting it as much as we've seen the 780G dominate most of the market. AMD's 780G and now 785G boards come with a good onboard solution (HD 3200 in the case of the 780G and HD4200 in the 785G) and also a HDMI port which makes them perfect for building cheap HTPCs that can play HD content

without breaking a sweat. Decent AMD 780G based boards can be found as cheap as Rs. 3,500.

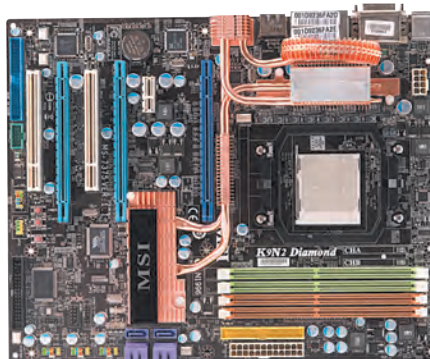
The Gigabyte MA785GM-US2H runs on AMD's 785 chipset. It uses the next generation of AMD's on-board graphics solutions the HD4200. It's weird to see that Gigabyte hasn't gone with a fully solid state capacitor board. Those in the processor region are solid capacitors while the rest are the traditional electrolytic ones. The heat sink on the Northbridge isn't very sturdy and wobbles from side to side a little bit. There's always that worry that it might not be

DVI and HDMI connectors. The HD4200 has a big role to play by trying to replace the older but popular HTPC solution chipset the 780G. This board is of the mATX form factor, so it's perfect for those building HTPCs.

Gigabyte's MA785GPM-UD2H runs on DDR3 and being an AMD 785 chipset, it's something people will soon start purchasing to build systems based on AMD processors. Board prices will soon drop along with DDR3 prices into a scenario similar to what it is today - AMD 780G boards running on DDR2 RAM. The audio chip is



MSI 790GX-G65



MSI K9N2 Diamond



making contact with the chip below. Most of the SATA and PATA connectors lie on the extreme right side of the board. The board comes with all VGA,

now upgraded to the Realtek ALC889A. The CMOS jumper is oddly placed between the two PCI slots. The board can be a little difficult to use. The only

HOW WE TESTED

For the AMD boards, a Phenom II x4 965 3.4 GHz processor was used. For Intel, it was the 3 GHz Intel Core 2 Duo QX9650 and the Core i7 975 (3.33 GHz) for x58 boards. We used Corsair Dominator dual channel kits for DDR2 and DDR3 modules; for the Core i7 Tri-channel kit we used Kingston memory. Western Digital's 300 GB Velociraptor HDDs were used on all test benches. The ASUS GeForce 9800GTX+ was used on the discrete graphics boards. All the systems were powered by Corsair's HX620 SMPS.

The staple diet of any motherboard or any kind of desktop performance test has to include PCMark; which itself contains a multitude of tests designed to stress each

component. It runs a wide variety of desktop applications and a few graphic tests to give a final score for the entire subsystem. This includes everything from the processor, memory, hard drive, graphics card, etc.

ScienceMark was used to get scores for the memory benchmarks. We expected to see large differences between DDR2 and DDR3 systems. The differences in between the DDR2 and DDR3 scores would be more interesting though.

For just pure processor and memory subsystem testing, a 3D rendering benchmark had to be used. Cinebench which is basically a dedicated benchmark utility based on the popular Cinema 4D 3D modelling program. Another

processor intensive test was WinRAR.

For a more real world test that we would use every day, we used a video compression program. DivX Converter was our choice because of its reliability. It's something we've used for a long time now. The standard 100 MB VOB file was to be converted and the time noted.

There was no simpler way to test the hard drive performance and the input/output subsystem than to do huge file copy tests on the hard drive. We used a single 8 GB file and then a folder with 8 GB of data in it. Times were recorded.

The graphics benchmarks were separated for boards with on-board graphics and the others. 3D Mark Vantage was used for

the motherboards tested with the dedicated graphics card. S.T.A.L.K.E.R Clear Sky and Crysis made up the real world games for this part of the test.

For boards with integrated graphics, it was older games such as Doom 3 and Company Heroes. 3D Mark Vantage was replaced by the older 3D Mark 2006. Games like Crysis and Stalker would bring these boards to a screeching halt.

With motherboards as always, there's a very tiny difference in performance between boards using the same chipset. Weights are given to sections of the test and as always performance was a small factor. The features and design of the board on the hand were given higher priority.